

## Exercise Set 10

17 November 2025

### 1 Abstract Interpretation II: Lattices

1. Are the following posets a lattice? a complete lattice?

- \*  $(\mathbb{N}, \leq)$
- \*  $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$
- \*  $(\mathcal{P}(\mathbb{Z}), \subseteq)$
- \*  $(E, \leq)$  where  $E$  is a finite

2. We define the set of intervals as

$$\mathcal{I} = \{[a, b], a, b \in \mathbb{Z}\} \cup \{(-\infty, a], a \in \mathbb{Z}\} \cup \{[a, +\infty), a \in \mathbb{Z}\} \cup \{(-\infty, +\infty)\}$$

Define an order on  $\mathcal{I}$  which represents the fact that an interval is less precise than another.

Express the *meet* and the *join* (i.e., given two intervals  $i$  and  $j$  what is  $\sup(i, j)$  and  $\inf(i, j)$  using the order defined before).

Prove that  $\mathcal{I}$  endowed with this order is a complete lattice

3. Let  $(E, \leq)$  and  $(F, \leq)$  be two complete lattices. Let  $f : E \rightarrow F$  be a function. We state

- \* if  $f$  is continuous, then it is monotonic (non-decreasing)
- \* the reverse does not hold in general (hint: consider for example  $f : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N})$  s.t.  $\forall X \subseteq \mathbb{N}$ , then  $f(X) = \mathbb{N}$  if  $X = \mathbb{N}$ ,  $f(X) = \emptyset$  otherwise)
- \* the reverse holds if we assume that  $E$  is finite

Prove the three previous assertions.

Now, let  $(G, \leq)$  be a third complete lattice and  $g : F \rightarrow G$  be a function. Prove that,

- \* if  $f$  and  $g$  are continuous, then  $g \circ f$  is also continuous.

4. Particular functions on signs and intervals.

- \* Consider the complete lattice of signs  $(\mathcal{S}, \leq)$  (with a single variable  $x$ ) and  $f_k : s \rightarrow s[x := x + k]$  for  $k \in \mathbb{N} \setminus \{0, 1\}$ .

Define  $f_k$  on  $\mathcal{S}$  and prove that  $f_k$  is continuous and monotonic.

- \* Consider the complete lattice of signs  $(\mathcal{S}, \leq)$  (with a single variable  $x$ ) and  $g_k : s \rightarrow s \cap [x > k]$  for  $k \in \mathbb{N}^*$ .  
Define  $g_k$  on  $\mathcal{S}$  and prove that  $g_k$  is continuous and monotonic.
  - \* Consider the complete lattice of intervals  $(\mathcal{I}, \leq)$  (with a single variable  $x$ ) and  $f_k : s \rightarrow s[x := x + k]$  for  $k \in \mathbb{N}$ .  
Define  $f_k$  on  $\mathcal{I}$  and prove that  $f_k$  is continuous and monotonic.
  - \* Consider the complete lattice of intervals  $(\mathcal{I}, \leq)$  (with a single variable  $x$ ) and  $g_k : s \rightarrow s \cap [x > k]$  for  $k \in \mathbb{N}^*$ .  
Define  $g_k$  on  $\mathcal{I}$  and prove that  $g_k$  is continuous and monotonic.
5. Let  $(E, \leq)$  and  $(F, \leq)$  be two complete lattices. We define  $\leq$  on  $E \times F$  as the conjunction of the comparisons on  $E$  and  $F$ :  $\forall (a, b), (c, d) \in E \times F, (a, b) \leq (c, d)$  if and only if  $a \leq c$  and  $b \leq d$ . Prove that
- \*  $\leq$  on  $E \times F$  is actually an order
  - \*  $(E \times F, \leq)$  is also a complete lattice.
6. Given the following pieces of code, draw the CFG, write the system of equations and solve them by applying Kleene's theorem

|                                                            |                                                                    |
|------------------------------------------------------------|--------------------------------------------------------------------|
| <pre> x := -5; while (x &lt; 0) do   x := x+1 done; </pre> | <pre> y := 3; x := 4; while (x &gt;= y) do   x := x-4 done; </pre> |
|------------------------------------------------------------|--------------------------------------------------------------------|

- \* Do it for the signs lattice
- \* Do it also for the intervals lattice

For the second program, we have two variables  $x$  and  $y$ . How do we define  $V(\alpha) \cap [x \geq y]$ ?

7. We want to define a Galois connection between the a concrete set  $\mathbb{Z}$  and the abstract set of the intervals. Define the  $\alpha$  and  $\gamma$  functions. Show that this is indeed a Galois connection
8. Let  $(E, \leq)$  and  $(F, \leq)$ ,  $\alpha : E \rightarrow F$ ,  $\gamma : F \rightarrow E$  be a Galois connection. Prove the following properties
- \*  $\forall x \in E, x \leq (\gamma \circ \alpha)(x)$
  - \*  $\forall y \in F, (\alpha \circ \gamma)(y) \leq y$
  - \*  $\alpha$  is monotonic (non-decreasing)
  - \*  $\gamma$  is monotonic (non-decreasing)

\*  $\alpha$  is Scott-continuous

9. Let  $(E, \leq)$  and  $(F, \leq)$  be two complete lattices. Let  $f : E \rightarrow F$  and  $g : E \rightarrow F$  be two functions and assume that both  $f$  and  $g$  are Scott-continuous. Prove that

\*  $\text{lfp}(f) \leq \text{lfp}(g)$